

Corneal melt and tectonic rescue

Part 2 | Peripheral ulceration may be systemic disease

Exclude infection while looking beyond the cornea.

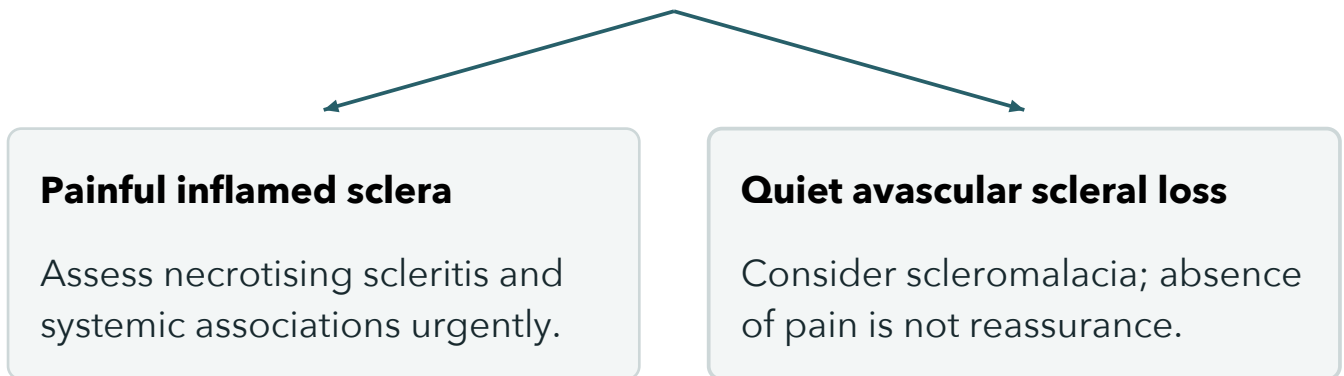
Assessment domain	High-value question
Corneal edge	Is there epithelial loss and progressive stromal thinning?
Adjacent sclera	Is necrotising inflammation present?
Systemic history	Is there vasculitis or inflammatory arthritis?
Microbiology	Could immunosuppression worsen an occult infection?

PUK can accompany systemic autoimmune disease and may signal serious vasculitis. Coordinate systemic assessment and anti-inflammatory treatment when an immune mechanism is established. Local protection alone does not control the systemic driver, while empiric immunosuppression without considering infection can accelerate tissue destruction.

Pain and redness do not fully grade scleral risk

Necrotising disease can be inflamed or deceptively quiet.

What surrounds the thinning?



Distinguish episcleritis from deeper scleral inflammation by examination and clinical context, not symptom intensity alone. Posterior scleritis can need ultrasound or other imaging. Infection must be considered, especially after surgery or trauma. A focal injection may reduce symptoms without establishing that a destructive process is safe.

Similar peripheral patterns require different treatment

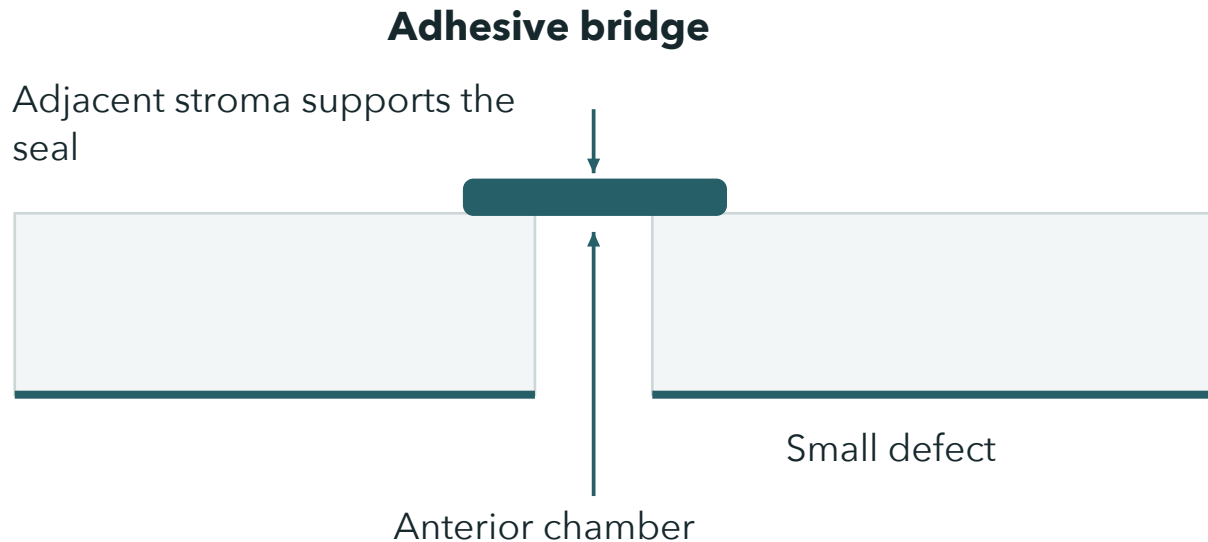
The relationship to the limbus and systemic findings matters.

Pattern	Mechanism to consider
Peripheral infiltrates with a clear limbal interval	Staphylococcal antigen-related marginal keratitis
Painful progressive peripheral ulcer with undermined edge	Mooren-type ulcer after exclusions
Ulceration with adjacent scleral disease	Systemic immune PUK or infection
Positive culture with tissue progression	Treat microbial disease first

Mooren ulcer is an exclusion diagnosis, without the systemic disease or scleral involvement expected in other destructive patterns. Do not infer sterility from the word peripheral. Lid disease supports marginal keratitis, but atypical ulceration or progression requires reassessment.

A temporary seal buys time

Geometry and adjacent tissue determine whether glue is plausible.



Original section: adhesive forms a bridge over a small defect; it does not regenerate stroma.

A small suitably shaped perforation may be stabilized with tissue adhesive and a protective lens under specialist care. A large, limbal or poorly supported defect may need another tectonic procedure. Continue treatment of infection or inflammation; successful cessation of leakage is only the first endpoint.

Tectonic and optical goals are different

Restore globe integrity before pursuing an ideal refractive result.

Immediate goal	Later goal
Seal leakage and preserve tissue	Improve optical clarity
Control infection or destructive inflammation	Choose rehabilitation after stability
Protect lids and tear coverage	Optimize contact lens or graft performance

A graft placed into uncontrolled inflammation or infection faces a hostile recipient environment. Urgent tissue replacement can nevertheless be necessary to preserve the eye. Explain the staged goals: structural rescue may leave irregular astigmatism or opacity, and later optical surgery requires renewed assessment of disease control.